St. Philomena's College (Autonomous), Mysore

PG Department of Studies and Research in Chemistry

Question {Revised Curriculum (LOCF) - 2020-22 Batch }

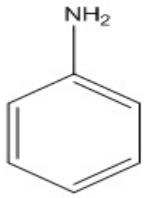
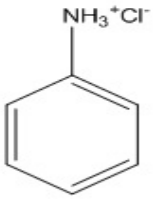
Second Year- Third Semester

Course Title:
MOLECULAR SYMMETRY AND SPECTROSCOPY, QP Code:85322

Unit	SI. No	Questions	Marks
1	1	Identify the symmetry elements present in N_2H_2 (trans structure)	02
1	2	Identify the symmetry elements present in Naphthalene	02
1	3	Identify the symmetry elements present in $POCl_3$	02
1	4	Identify the point group to which the following molecules belong PF_5, PCl_3	02
1	5	Write down all the symmetry elements of methane	02
1	6	Among the following, why does only XeF_4 contain a 4-fold principal rotation axis: CF_4, SF_4, $[BF_4]$- and XeF_4?	02
1	7	N_2O_4 is planar. Show that it possesses three planes of symmetry	02
1	8	Why does CO_2 possess an inversion center, but NO_2 does not?	02
1	9	Show that the planes of symmetry in benzene are one σh, three σv, and three σd.	02
1	10	How many planes of symmetry do (a) $F_2C=O$, (b) $ClFC=O$ and c) $[HCO_2]$- possess?	02
1	11	What symmetry operators are lost in going from NH_3 to NH_2Cl?	02
1	12	Compare the symmetry operators possessed by NH_3, NH_2Cl, $NHCl_2$, and NCl_3.	02
1	13	Draw a diagram to show the symmetry operators of $NClF_2$.	02
1	14	Show that BF_3 and $F_2C=O$ have the following symmetry elements in common E, two mirror planes, one C_2.	02
1	15	How do the symmetry elements of ClF_3 and BF_3 differ?	02
1	16	What symmetry elements do BCl_3 and PCl_3 (a) have in common and (b) not have in common?	02
1	17	Draw the structure of SO_2 and identify its symmetry, properties.	02
1	18	Which of the following molecules contains a C_4 axis and a σ_h plane: CCl_4; $[ICl_4]^-$;sulphate; SiF_4; XeF_4?	02
1	19	How many mirror planes do each of the following molecules contain: a)SF_4; b) H_2S; c)SF_6; d) SOF_4; e) SO_2; f) SO_3 ?	02
1	20	Which of the following species contain inversion centers? a) BF3; b) SiF4; c) XeF4; d) PF5; e) XeF_5; f) SF_6; g)C_2F_4; f) SF_6	02
1	21	Write the signs of 8 octants in a cube of the x,y,z molecular planes	02
1	22	List the groups which are the subgroups of D_{6h} and C_{3v}	02
1	23	List the groups which are the subgroups of C_{2v} and T_d	02
1	24	List the groups which are the subgroups of C_{2v} and O_h	02
1	25	List the groups which are the subgroups C_{4v} and C_{2h}	02

1	26	List the groups which are the subgroups D_{2h} and D_{2d}	02
1	27	List the groups which are the subgroups C_i and C_s	02
1	28	Identify the point group to which the following molecules belong PCl_3F_2, SF_4BrCl	02
1	29	Identify the point group to which the following molecules belong Cis-butadiene, transbutadiene.	02
1	30	Identify the point group to which the following molecules belong a) hexatriene b) cis-dichloroethene,	02
1	31	Identify the point group to which the following molecules belong boat form cyclohexane, NH_2F	02
1	32	Identify the point group to which the following molecules belong carbonate, sulphate ion	02
1	33	Identify the point group to which the following molecules belong monoclorobenzene, Ortho-dichlorobenzene	02
1	34	Identify the point group to which the following molecules belong Mono chloroacetylene, dichloroethylene	02
1	35	Identify the point group to which the following molecules belong hydrogen peroxide, formaldehyde	02
1	36	Identify the point group to which the following molecules belong CH_3F, CH_2F_2	02
1	37	To what point group does $POCl_3$ belong?	02
1	38	Assign the point groups to ammonium and ammonia molecular systems	02
1	39	To which point group does NF_3 belong?	02
1	40	The point group of $[AuCl_2]^-$ is D∞h What shape is this ion?	02
1	41	Determine the point group of SF_5Cl.	02
1	42	Assign a point group to each member in the series (a) CCl_4, (b) CCl_3F, (c) CCl_2F2, (d) $CClF_3$ and (e) CF_4.	02
1	43	Deduce the point group of SF4. Is SOF_4 in the same point group?	02
1	44	Which of the following point groups possesses the highest number of symmetry elements: (a) O_h; (b) T_d; (c) I_t?	02
1	45	$[PtCl_4]^{2-}$ has a center of symmetry but $[CoCl_4]^{2-}$ does not. One is square planar and the other is tetrahedral which is which?	02
1	46	CS_2 and HCN are both linear. Explain why CS_2 possesses a center of symmetry whereas HCN does not.	02
1	47	Show that operations inversion and rotation commute	02
1	48	Show the abelian nature of C_s point group	02
1	49	Show the abelian nature of C_i point group	02
1	50	Show the abelian nature of S_4 point group	02
1	51	Show the abelian nature of C_4 point group	02
1	52	$[PtCl_4]^{2-}$ is square planar to what rotation operation is equivalent?	02
1	53	Explain what is meant by an ∞-fold axis of rotation.	02
1	54	Write the group multiplication table for C_2	02

1	55	The following are sets of numbers. Which of them are from a group? The mode of combination is given in parenthesis a) 1,-1, 0 (addition) b) 2,-2,0 (addition)	02
1	56	The following are sets of numbers. Which of them from a group? The mode of combination is given in parenthesis a) i,-1,+1, -i (multiplication) b) i,-1,+1, -i (addition)	02
1	57	P.T The numbers ±1 and ±i form a group under multiplication	02
1	58	What do you mean by reducible and irreducible representation	02
1	59	Why do CO_2 and SO_2 have a different number of degrees of vibrational freedom?	02
1	60	How many degrees of vibrational freedom does each of the following possess: $SiCl_4$, BrF_3, $POCl_3$?	02
1	61	How many IR active vibrational modes does CS_2 possess, and why? Hint: CS_2 is isostructural with CO_2.	02
1	62	Determine the number of degrees of vibrational freedom for each of the following: (a) SO_2; (b) SiH_4; (c) HCN; (d) H_2O; (e) BF_3.	02
1	63	How many normal modes of vibration are IR active for (a) H_2O, (b) SiF_4, (c) PCl_3, (d) $AlCl_3$, (e) CS_2 and (f) HCN?	02
1	64	Write the elements of symmetry and GMT for deuterated water	02
1	65	Write the elements of symmetry and GMT for HOD	02
1	66	Write the elements of symmetry and GMT for HOCl	02
1	67	Write the elements of symmetry and GMT for HOBr	02
1	68	Write the elements of symmetry and GMT for HOI	02
1	69	Write the elements of symmetry and GMT for NH2Cl	02
1	70	Write the elements of symmetry and GMT for NHCl2	02
3	71	Calculate the wavenumber of radiation whose wavelength is 4 μm.	02
3	72	Calculate the frequency of radiation whose wavelength is 400 nm. Express this wavelength in wavenumber	02
3	73	Calculate the frequency of the radiation in hertz of wavelength 350 nm	02
3	74	Calculate the wavenumber of the radiation if the frequency is 2.06×10^{14} Hz	02
3	75	Calculate the wavelength in 'nm' if the frequency of the strong yellow line in the spectrum is 5.09×10^4 s^{-1}	02
3	76	Determine the wavelength of an electron moving with one third the speed of light	02
3	77	Calculate the wavelength, wavenumber, and frequency of radiation having an energy of 1.5 eV	02
3	78	For infrared radiation of 5.00 μm, what is the wavenumber in cm^{-1}	02
3	79	Calculate the quantum of energy, E in joules, J mol^{-1}, cal mol^{-1}, and eV for a wavelength of 200 nm	02
3	80	Why is ethanol is a good solvent in ultraviolet	02
3	81	What are auxochromes and chromophores? Give an example for each.	02
3	82	Explain why it is necessary to use quartz cuvettes while recording UV spectra.	02

3	83	Which of the following compounds have a higher absorption maximum 1  2 	02
3	84	Which of the following would show the ESR spectrum? i) Methyl radical ii) Nitrogen molecule iii) Carbon dioxide iv) Benzene radical	02
3	85	A specimen of ZnO shows a strong ESR line at the frequency of 9.45 GHz. Calculate the g-value of the line.	02
3	86	The benzene anion has g = 2.0025. At what magnetic induction B_z would its ESR spectral line be centered at frequency 9.350 GHz?	02
3	87	Predict the number of components and their intensities in the ESR spectrum of benzene radicals.	02
3	88	How many ESR spectral lines are observed if an unpaired electron, delocalized on two non-equivalent protons, is placed in a magnetic field	02
3	89	An ESR spectrum shows only three lines. Explain how it could arise from Hyperfine coupling to two nuclei with spin = ½ Hyperfine coupling to two nuclei with spin = 1	02
3	90	The ESR spectrum of a radical with a single magnetic nucleus is split into four lines of equal intensity. What is the spin of the nucleus?	02
3	91	How many lines would you expect in the ESR spectra of bis(acetylacetonate) copper(II) if I for copper is 3/2?	02
3	92	Predict the ESR spectrum pattern for $NO(SO_3)_2^{2-}$ radical if I=1 for nitrogen which is the only nucleus having a spin	02
3	93	In the ESR spectrum of $K_3Mo(CN)_8$, 25% isotopes of Mo show spin I=5/2. Predict the pattern of lines around a central line from 75% non-magnetic isotope	02
3	94	ESR is observed for hydrogen atoms with an instrument operating at 9.5 GHz. If the g value for the electron atom is 2.0026, what is the magnetic field applied? Bohr magneton= 9.274×10^{-24} JT^{-1}	02
3	95	A free electron is placed in a magnetic field of strength 1.3 T. Calculate the resonance frequency if g= 2.0023	02
3	96	How do crystal field and spin-orbit coupling affect the value of g?	02

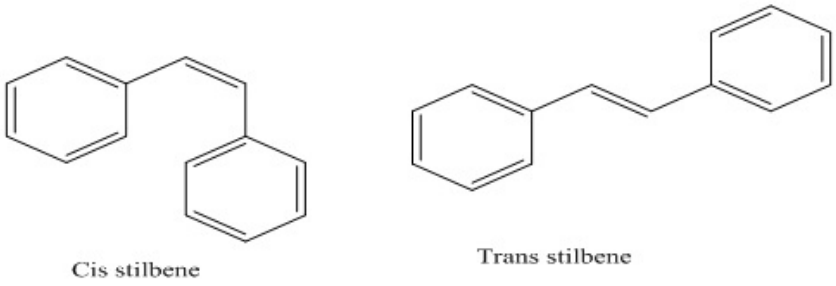
3	97	Why the line width of the ESR signal is larger than that of the NMR signal? Why is the ESR signal is presented in derivative form	02
3	98	What is zero-field splitting? Explain with suitable example	02
2	99	What is the equation of rotational constant, B?	02
2	100	Which of the following molecule will show rotational spectra? O = C = O, HF, N_2, ^{16}O= C= ^{17}O	02
2	101	Give the equation of Rotational constant and moment of inertia.	02
2	102	Which of the following molecule will show rotational spectra? O = C = O, HF, N_2	02
2	103	Calculate the energy in terms of v of the energy level corresponding to J = 7	02
2	104	What is the selection rule for a rigid diatomic molecule to show a rotational spectrum?	02
2	105	Give the expression for the energy for the rigid diatomic rotor	02
2	106	State why a molecule needs to have a permanent dipole moment to be microwave active?	02
2	107	Write the selection rule for rotation of the non-rigid rotator.	02
2	108	Write down one example of symmetric top and spherical top molecules.	02
2	109	Write the condition for the moment of inertia for linear molecules.	02
2	110	Write the condition for the moment of inertia for symmetric top molecules.	02
2	111	Write the condition for the moment of inertia for prolate molecules.	02
2	112	Write the condition for the moment of inertia for oblate molecules.	02
2	113	Write the expression for moment of inertia for polyatomic molecules.	02
2	114	Write the condition for the moment of inertia for asymmetric top molecules.	02
2	115	What is meant by zero-point energy?	02
2	116	What is the force constant? Write the units.	02
2	117	Write the equation for the energy of an anharmonic oscillator	02
2	118	write the expression for fundamental frequency in terms of force constant.	02
2	119	What is Hooke's law?	02
2	120	What is the force constant? What is its unit?	02
2	121	What is the force constant? What is its significance? Write the units.	02
2	122	What is the stark effect? Outline the importance of stark effect studies in microwave spectroscopy	02
2	123	Write the possible vibrational modes of CO_2 molecules and mention their IR activity	02
2	124	Why carbon dioxide has two absorption peaks in its IR Absorption spectrum.	02
2	125	Give the equations of fundamental absorption and first overtone.	02
2	126	Find the number of vibration modes of the non-linear molecule with 7 atoms	02
2	127	Write the different modes of vibrations of carbon dioxide.	02
2	128	Write the possible vibrational modes of CO_2 molecules and mention their IR activity	02

2	129	How fundamental bands are different from overtone bands	02
2	130	State the total number of vibrational modes in triatomic linear and symmetric molecules.	02
2	131	Calculate the theoretical number of vibrational degree of freedom in (a)Benzene (b) carbon dioxide	02
2	132	Calculate the theoretical number of vibrational degrees of freedom	02
2	133	How fundamental bands are different from overtone bands ?	02
2	134	What is Raman scattering?	02
2	135	Define stark effect in rotation spectra.	02
2	136	Write down the Stark effect	02
2	137	Write down the applications of microwave spectroscopy.	02
2	138	Calculate the theoretical number of vibrational degree of freedom in (a)Benzene (b) SO_2	02
2	139	Calculate the energy in terms of $\bar{\upsilon}$ of the energy level corresponding to J = 7.	02
2	140	Write the condition for the moment of inertia for asymmetric top molecules with an example	02
2	141	write the equation for the energy for the rigid diatomic rotor	02
2	142	Give the schematic representation of the energy changes associated with the excitation for Rayleigh and Raman scattering	02
2	143	Define elastic and inelastic collisions.	02
2	144	What is Raman shift? The Raman shift positions of Stokes and anti-Stokes lines are equal but opposite. Comment.	02
2	145	Calculate the wavenumber (in cm^{-1}) of the radiation having a wavelength of 488 nm.	02
2	146	State the rule of Mutual Exclusion.	02
2	147	Enlist the applications of Raman spectroscopy	02
2	148	State mutual exclusion principle activities for deciding IR and Raman	02
1	149	Show that C_{2z}, C_{2x}, C_{2y} commute, and the product of any two is equivalent to the third	05
1	150	Prove that The readings 1,2,3,…..12 on the dial of a clock form a group under multiplication	05
1	151	Draw the structure of $[XeF_5]^-$, on the diagram mark the C_5 axis. The molecule contains five C_2 axes. Where are these axes?	05
1	152	Determine the point group of trans-N_2F_2 using the flowchart.	05
1	153	Determine the point group of PF_5 using the flowchart.	05
1	154	Show that BF_3 belongs to the D_{3h} point group using a flow chart	05
1	155	Show that OF_2 belongs to the C_{2v} point group using a flow chart	05
1	156	The flowchart shows that $CHCl_3$ possesses C_{3v} symmetry, but that CCl_4 belongs to the T_d point group.	05

1	157	Write the group multiplication table for a water molecule	05
1	158	Write the group multiplication table for ammonia molecule	05
1	159	Construct the matrices for symmetry operations in the C_{2v} point group using H-1s orbitals and O-2pz orbital of a water molecule (assume the z-axis to be C_2 axis). Write down the characters for this reducible representation and decompose them into its irreducible representation	05
1	160	Give the GOT and its consequenses for C_{2v} character table	05
1	161	Give the GOT and its consequenses for C_{3v} character table	05
1	162	Give the GOT and its consequences for the C_{2h} character table	05
1	163	Find out the symmetries of the fundamentals of sulphate and carbonate. Comment on their infrared and Raman activities	05
1	164	Prove that in a centrosymmetric molecule (the symmetry element i is present) no IR active normal mode can also Raman active and vice versa	05
1	165	Verify the symmetry designations of the normal coordinates of the water molecule	05
1	166	Find the symmetry coordinates which can be used to construct the normal coordinates of the trans N_2F_2 molecule.	05
1	167	Use VSEPR theory to help you, draw the structures of CF_4, XeF_4, and SF_4. Assign a point group to each molecule. Show that the number of degrees of vibrational freedom is independent of molecular symmetry.	05
1	168	The IR spectrum of $SnCl_2$ exhibits absorptions at 352, 334, and 120 cm^{-1} what shape do these data suggest for the molecule, and is this result consistent with VSEPR theory?	05
1	169	The vibrational modes of XeF_2 are at 555, 515, 213 cm^{-1} but only two are IR active. Explain why this is consistent with XeF_2 having a linear structure.	05
1	170	The IR spectrum of SF2 has absorptions at 838, 813, and 357 cm^{-1}. Explain why these data are consistent with SF_2 belonging to the C_{2v} rather than the $D_{\infty h}$ point group.	05
1	171	To what point group does F_2O belong? Explain why the vibrational modes at 928, 831, and 461 cm^{-1} are all IR active.	05
1	172	The IR spectrum of BrF_3 in an argon matrix shows six absorptions Explain why this observation confirms that BrF_3 cannot have C_{3v} symmetry.	05
1	173	The IR spectrum of BF_3 shows absorptions at 480, 691, and 1449 cm-1. Use these data to decide whether BF_3 has C_{3v} or D_{3h} symmetry.	05
1	174	The IR spectrum of gaseous ZrI_4 shows absorptions at 55 and 254 cm^{-1} Explain why this observation is consistent with molecules of ZrI_4 having T_d symmetry.	05
1	175	The $[PdCl_4]^{2-}$ ion gives rise to three absorptions in its IR spectrum (150, 321, 161 cm^{-1}) Rationalize why this provides evidence for a D_{4h} rather than T_d structure.	05
1	176	Discuss the Consequences of GOT for water molecular system	05
1	177	Discuss the Consequences of GOT for the OF2 molecular system	05

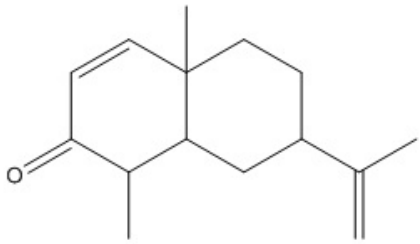
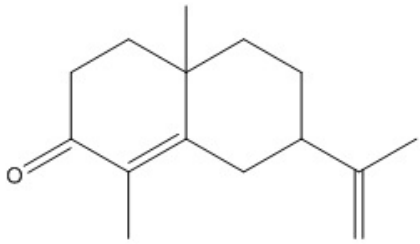
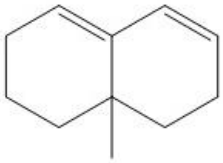
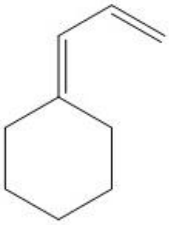
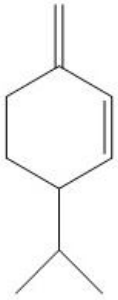
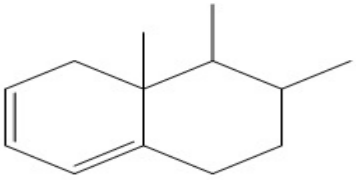
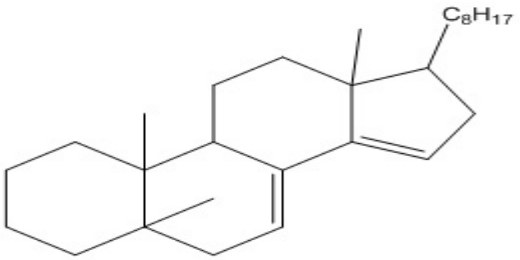
1	178	Discuss the Consequences of GOT for BF2Cl molecular system	05
1	179	Discuss the Consequences of GOT for cis-dichoroethylene	05
1	180	Discuss the Consequences of GOT for pyridine	05
1	181	Discuss the Consequences of GOT for formaldehyde	05
1	182	Discuss the Consequences of GOT for ClF3	05
1	183	Discuss the Consequences of GOT for cis-N2F2	05
1	184	Discuss the Consequences of GOT for dichloromethane	05
1	185	Discuss the Consequences of GOT for m-dichlorobenzene	05
1	186	Discuss the Consequences of GOT for o-dichlorobenzene	05
1	187	Discuss the Consequences of GOT for selenium di oxide	05
1	188	Discuss the Consequences of GOT for xenon trioxide	05
1	189	Discuss the Consequences of GOT for ammonia	05
1	190	Discuss the Consequences of GOT for phosphorus oxychloride	05
1	191	Discuss the Consequences of GOT for transdichloroethylene	05
1	192	Discuss the Consequences of GOT for transdifluoroethylene	05
1	193	Discuss the Consequences of GOT for Trans N2F2	05
1	194	Explain the IR and Raman activity of vibrations for a water molecule	05
1	195	Explain the IR and Raman activity of vibrations for the OF2 molecular system	05
1	196	Explain the IR and Raman activity of vibrations for the BF2Cl molecular system	05
1	197	Explain the IR and Raman activity of vibrations for cis-dichloroethylene	05
1	198	Explain the IR and Raman activity of vibrations for pyridine	05
1	199	Explain the IR and Raman activity of vibrations for formaldehyde	05
1	200	Explain the IR and Raman activity of vibrations for ClF3	05
1	201	Explain the IR and Raman activity of vibrations for cis-N2F2	05
1	202	Explain the IR and Raman activity of vibrations for dichloromethane	05
1	203	Explain the IR and Raman activity of vibrations form-dichlorobenzene	05
1	204	Explain the IR and Raman activity of vibrations for o-dichlorobenzene	05
1	205	Explain the IR and Raman activity of vibrations for selenium dioxide	05
1	206	Explain the IR and Raman activity of vibrations for xenon trioxide	05
1	207	Explain the IR and Raman activity of vibrations for ammonia	05
1	208	Explain the IR and Raman activity of vibrations for phosphorus oxychloride	05
1	209	Explain the IR and Raman activity of vibrations for transdichloroethylene	05
1	210	Explain the IR and Raman activity of vibrations for transdifluoroethylene	05
1	211	Explain the IR and Raman activity of vibrations for Trans N2F2	05
3	212	Why there is no absorption due to n to σ^* transition in the spectrum of trimethylamine in the acidic solution	05
3	213	The absorption maximum for methyl chloride is 173 nm while for methyl iodide it is 259 nm. Why?	05
3	214	Why does hydrogen bonding shifts the absorption to shorter wavelengths in alcohols and amines?	05

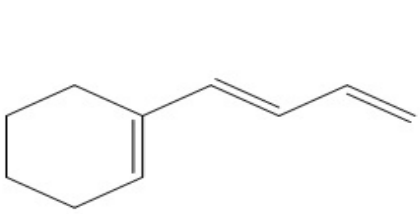
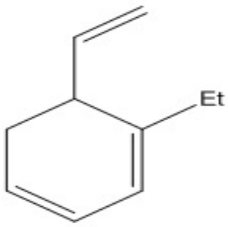
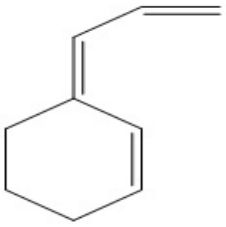
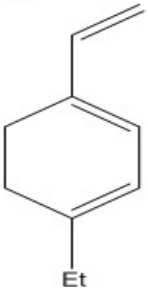
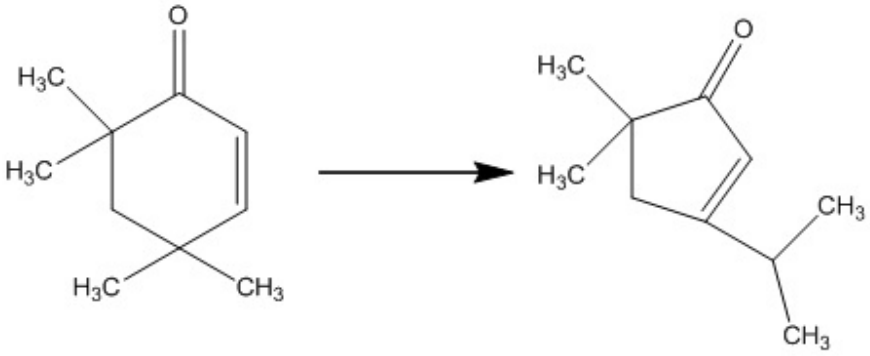
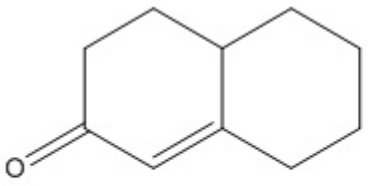
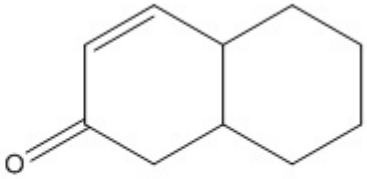
3	215	Amines absorb at a higher wavelength than alcohols. Why?	05
3	216	Name the electronic transition possible when UV light is absorbed	05
3	217	What determines the wavelength of UV absorption by organic compound	**05**
3	218	Prove that A = 2-log T	05
3	219	Why do most photometers and spectrophotometers do not give a linear readout in transmittance rather a linear readout in absorbance	05
3	220	The molar absorptivity of a particular solute is 2.1×10^4. Calculate the transmittance through a cuvette with a 5.0 cm light path for a 2.0 μM solution.	05
3	221	What types of electronic transition are possible for each of the following compounds Dimethyl ether and methyl vinyl ether	**05**
3	222	What types of electronic transition are possible for each of the following compounds Triethylamine and cyclohexane	05
3	223	What types of electronic transition are possible for each of the following compounds Acetaldehyde and cyclopentene	05
3	224	Calculate the absorbance and transmittance of a 0.002 M solution of a substance having a molar absorption coefficient of 460 M^{-1} cm^{-1}. The measurement is made by using a curvature of 1cm path length.	05
3	225	The UV-Vis spectra in the solution phase have a band structure whereas in the vapor phase we get sharp lines. Explain	**05**
3	226	The intensity of the light beam is reduced by 50% on passing through 1.0 cm of absorbing medium. What will be the reduction after going through 2.0 cm of the same medium?	05
3	227	What would be the wavelength of the absorption maximum of the certain compound CH_3CH_2=CHC(CH_3)=O using the Woodward-Fieser rule	05
3	228	The intensity of the light beam is reduced by 50% on passing through 1.0 cm of absorbing medium. What will be the reduction after going through 2.5 cm of the same medium?	05
3	229	Predict the transition involved in a) methyl iodide b) methyl iodide c) methanol d) trimethylamine	**05**
3	230	The position of absorption of acetone shifts in different solvents is 279 nm in hexane, 272 nm in ethanol, and 264.5 nm in water. Why?	05
3	231	The C=C generally absorbs an absorption band at about 180-190 nm while the C=O group in aldehydes and ketones show absorption maximum near 270-290 nm. Why?	05
3	232	The bands due to due to n to π^* transition in amine disappear in acid solution. Why?	05
3	233	Biphenyl shows the following ultraviolet absorption data. In its 2,2'-dimethyl derivative, the absorption pattern becomes almost similar to *O*-xylene. Why?	**05**
3	234	Why does a polar solvent shift n to π^* transition to a shorter wavelength and π to π^* transition to a longer wavelength?	05
3	235	Explain why spectra of solutions in the UV-Vis regions tend to consist of a few	05

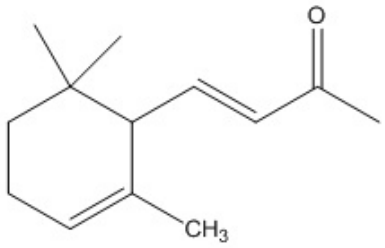
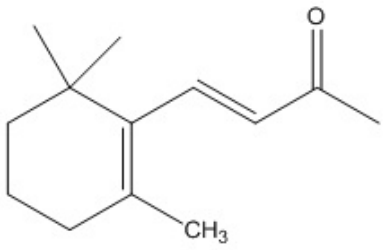
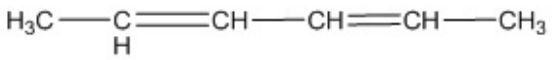
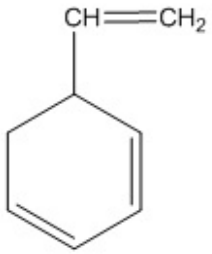
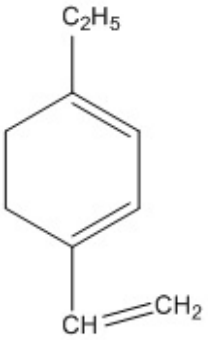
		broad peaks whereas the same solution measured in IR show many sharp peaks	
3	236	Explain why the UV spectrum of benzene vapor contains many sharp whereas the same spectrum of benzene solution in alcohol lacks fine detail and consists of only broad peaks	05
3	237	What is the absorbance of a solution containing 3 ppm if transmittance was 65% in 10 mm cuvette	05
3	238	Calculate the molar concentration and absorptivity of a certain compound (10 ppm) which had an absorbance of 0.650 at 310 nm, while an unknown compound (Molecular mass = 272.8) had an absorbance of 0.930	**05**
3	239	Calculate the molar absorptivity of a complex of iron with a 1,10-phenanthroline complex. If Fe (II) solution (4.7 ppm) when treated with reagent (made up to 100 ml) gave an absorbance of 0.467 at 510 nm	05
3	240	Calculate the concentration of the unknown solution of potassium dichromate if the transmittance measured was 75%. When one ml of 0.015 N potassium dichromate was added to the 5 ml cell the reading changed to 65% T.	05
3	241	How will you distinguish between cis-stilbene and trans-stilbene by the use of UV-Vis spectroscopy  Cis stilbene Trans stilbene	05
3	242	Mesityloxide, a condensation product of acetone is known to consist of two isomers (I) and (II) shown below. One isomer exhibits a maximum at 235 nm with ε= 12000 while the other shows no high-intensity absorption above 220 nm. Which of the two isomers absorbs at 235 nm? and Why?  I II	05
3	243	The UV Spectrum of acetone shows absorption bands at 166, 189, and 279 nm. What type of transition is responsible for each of these bands?	05
3	244	Chloromethane has an absorption maximum at 172 nm, bromomethane shows absorption at 204 nm and iodomethane shows a band at 258 nm. What	05

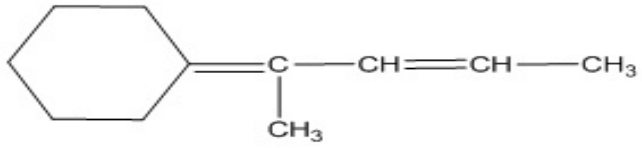
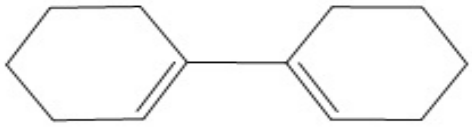
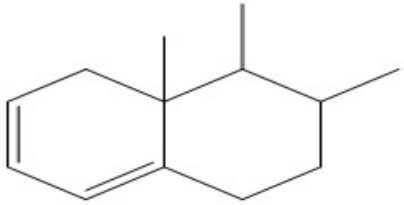
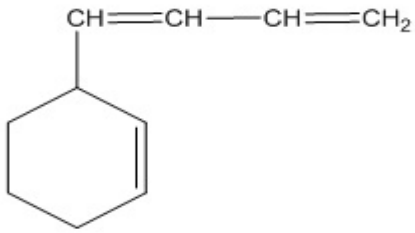
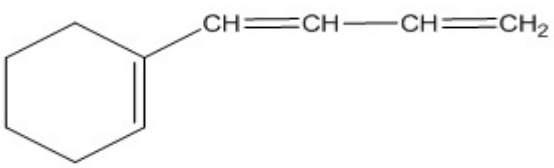
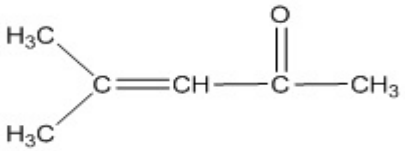
		type of transition is responsible for each band? How can the trend of absorption be explained?	
3	245	Explain different types of electronic transitions involved in CH_3CONH_2 with the help of a molecular orbital energy level diagram	05
3	246	The molar absorptivity of a substance is 1.5×10^4 cm^{-1} mol^{-1} dm^3. Calculate the transmittance through a cuvette of path length 50 mm containing 2×10^{-6} mol dm^{-3} solution of the substance.	05
3	247	Explain the effect of solvents on the π-π^* and n-π^* transitions	**05**
3	248	Discuss the bathochromic shift of conjugated dienes based on HOMO-LUMO theory	05
3	249	Explain how UV-Vis spectra can be used to differentiate between ethylene and butadiene by HOMO-LUMO theory	05
3	250	Discuss briefly different types of absorption bands in UV-Vis spectra.	05
3	251	What is the magnitude of molar absorptivity if a certain compound had a molecular mass = 92 if 0.015% solution by weight had 27% T with 10 mm cells?	05
3	252	What should be the magnitude of molar absorptivity if a solution was prepared by dissolving 0.1298 g of compound (molecular mass = 214.3) in one liter of solution. The absorbance of the new solution at 255 nm was 0.824?	**05**
3	253	A solution containing 8.75 ppm $KMnO_4$ has a transmittance of 0.743 in a 1.00 cm cell at 520 nm. Calculate the molar absorptivity of $KMnO_4$	**05**
3	254	Beryllium(II) (atomic mass : 9.01) forms a complex with acetylacetone (166.2 g/mol). Calculate the molar absorptivity of the complex, given that a 1.34 ppm solution has a transmittance of 55.7% when measured in a 1.00 cm cell at 295 nm, the wavelength of maximum absorption.	05
3	255	A 5.00 ml aliquot of a solution contains 5.94 ppm iron(III) is treated with an appropriate excess of KSCN and is diluted to 50.0 ml. what is the absorbance of the resulting solution at 580 nm in a 2.50 cm cell (ε=7.00×10^3 L mol^{-1} cm^{-1})	05
3	256	When 5 mg of a certain amine picrate is dissolved in 100 ml of ethanol, its absorbance is measured in a 2.00 cm cell at 380 nm as 0.415. What is the molecular weight of the amine picrate? Given: ε= 13400 M^{-1} cm^{-1}	05
3	257	Calculate the concentration in μg ml^{-1} of a solution of organic compound (mol mass 211.0) in 0.11 M HCl giving an absorption at its λ_{max} 281 nm of 0.612 in 4 cm cell. The molar absorptivity at 281 nm is 5372.	05
3	258	In the determination of acetone in biological fluids, a water blank instead of a reagent blank was used. A 2.5 mg% standard gave an absorbance of 0.81 and unknown gave 0.6. In a separate determination, it was found that the reagent blank gave an absorbance of 0.09 when read against the water blank. Calculate the concentration of the acetone in the unknown.	**05**
3	259	The absorption spectrum for titanium peroxide complexion in perchloric acid showed a maximum of 400 nm. The absorbance of 32.0 μg/ml of titanium gave	05

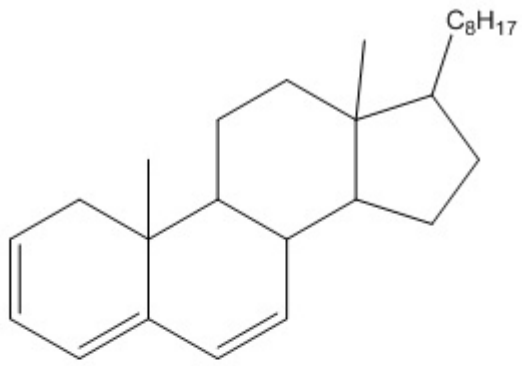
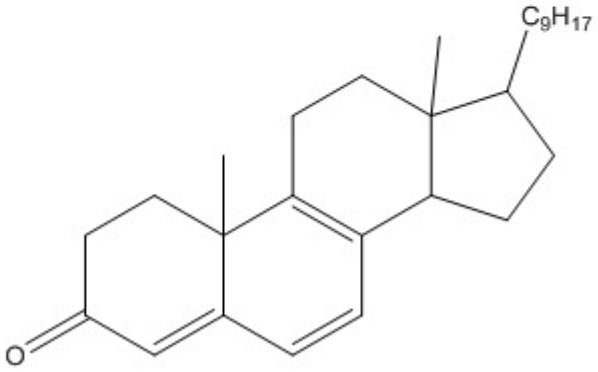
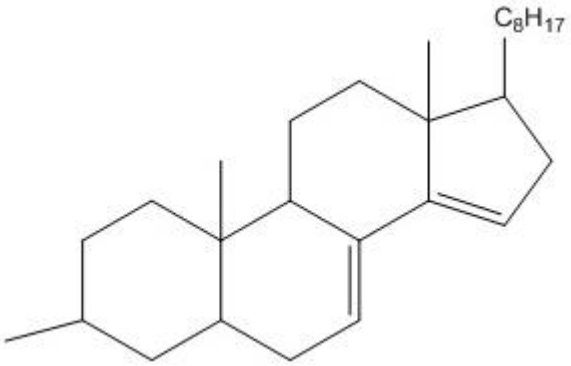
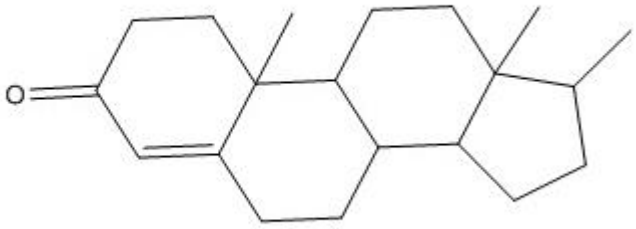
		an absorbance of 0.456. An unknown solution treated identically gave an absorbance of 0.501. assuming identical cells, find out the concentration of the unknown.	
3	260	Draw the UV-Vis energy level transitions for acetaldehyde, formaldehyde, and acrylaldehyde.	05
3	261	The wavenumber of absorption due to carbonyl increases in acid halides. Discuss the change in the absorption with a suitable explanation	05
3	262	Discuss in detail ESR peaks would you get for sodium atom (I=3/2)	05
3	263	Discuss in detail the ESR spectrum of 1,4-benzoquinone radical.	**05**
3	264	Explain in detail ESR spectrum of the following radicals (CF_2H and $^{13}CF_2H$)	05
3	265	Discuss the principle behind ESR spectroscopy. What type of species are studied by this technique.	05
3	266	What is Lande splitting factor (g)? Explain the factors affecting the g value?	05
3	267	Discuss in detail the ESR spectrum of the following species Methyl radical Benzene radical	05
3	268	Discuss in detail the ESR spectrum of the following species Naphthyl radical CH_2F radical $^{13}CF_2H$ radical	05
3	269	What do you mean by hyperfine structure? How do they arise in ESR spectra?	**05**
3	270	Describe the experimental technique for recording ESR spectrum	05
3	271	Describe the instrumentation and recording of the ESR spectrum. Why signals are recorded in terms of derivative form	05
3	272	Predict based on the HMO theory, the wavelength of the π to π^* transition observed in the UV spectra of the conjugated linear polyenes, butadiene, hexatriene, and octatetraene	05
3	273	Discuss the general application of UV-Vis spectroscopy in organic chemistry.	05
3	274	Discuss any three applications of UV-Vis spectroscopy	**05**
	275	α-hyper one exists as	05

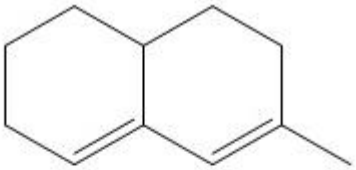
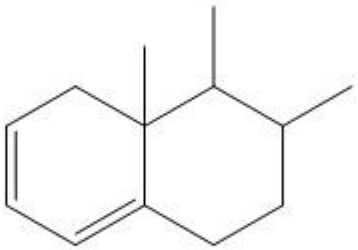
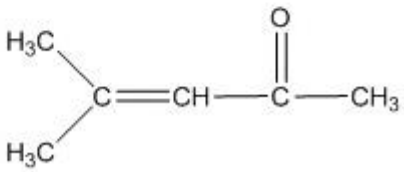
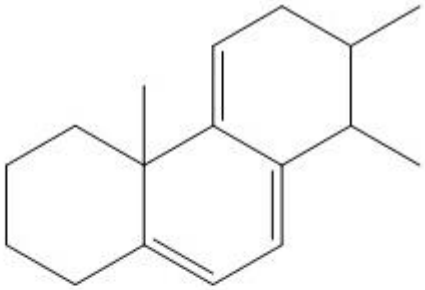
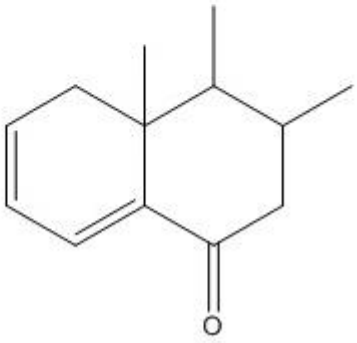
		  Using Woodward rules decide which among the above structures is consistent with the observed value of 252 nm	
3	276	Following the Woodward Fieser rules, calculate the absorption maximum for each of the following compounds a.  c.  b.  d. 	05
3	277	Using Woodward's rules, predict the wavelength of maximum absorption for each of the following compounds a.  b. 	05
3	278	Explain how the ultraviolet spectrum can be used to decide between the following isomeric systems	05

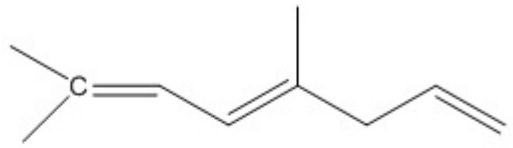
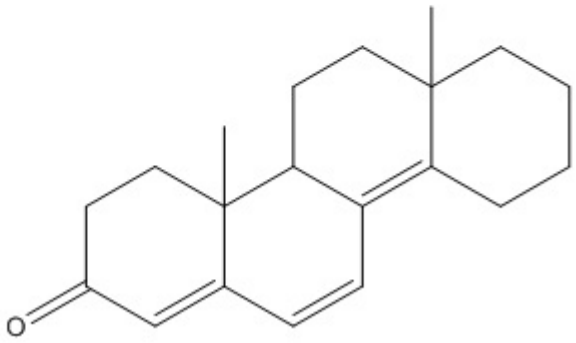
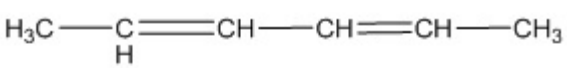
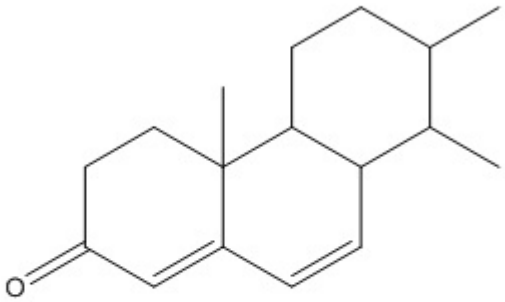
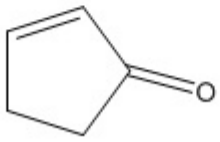
		 a  c  b  d	
3	279	Predict the UV maximum for the reactant and product of the following photochemical reaction: 	**05**
3	280	Predict and explain whether UV Visible spectroscopy can be used to distinguish between the following pairs of compounds. If possible support your answer with calculations  a  b	05

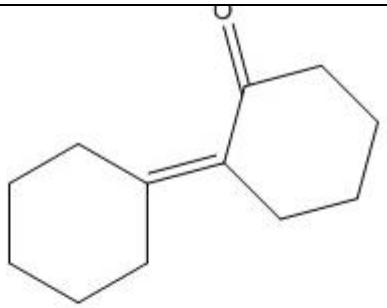
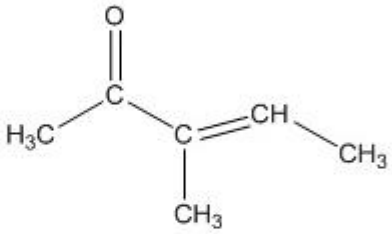
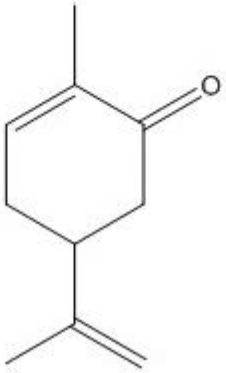
3	281	Predict and explain whether UV Visible spectroscopy can be used to distinguish between the following pairs of compounds. If possible support your answer with calculations  a  b	05
3	282	Calculate the absorption maximum for the following compounds by Woodward-Fieser rules a.  b.  c. 	05
3	283	Calculate the absorption maximum for the following compounds by Woodward-Fieser rules	05

		a.  b.  c. 	
3	284	Calculate the absorption maximum for the following compounds by Woodward-Fieser rules a.  b.  c. 	**05**
3	285	Calculate the absorption maximum for the following compounds by Woodward-Fieser rules	05

		a.  b. 	
3	286	Calculate the absorption maximum for the following compounds by Woodward-Fieser rules a.  b. 	05
3	287	Calculate the absorption maximum for the following compounds by	05

		Woodward-Fieser rules a.  b.  c. 	
3	288	Calculate the absorption maximum for the following compounds by Woodward-Fieser rules a.  b. 	05
3	289	Calculate the absorption maximum for the following compounds by	**05**

		Woodward-Fieser rules a.  b. 	
3	290	Calculate the absorption maximum for the following compounds by Woodward-Fieser rules a.  b.  c. 	05
3	291	Calculate the absorption maximum for the following compounds by Woodward-Fieser rules	05

		a.  b.  c. 	
3	292	At 580 nm, the wavelength of its maximum absorption, the complex $FeSC^{N2+}$ has a molar absorptivity of 7.00×10^3 L mo^{l-1} cm^{-1}, calculate The absorbance of a 3.75×10^{-5} solution of the complex at 580 nm in a 1.00 cm cell The absorbance of a solution in which the concentration of the complex is twice that in a) The transmittance of the solutions described in a) and b) The absorbance of a solution that has half the transmittance of that described in (a)	05
3	293	A solution containing the complex formed between Bi(III) and thiourea has a molar absorptivity of 9.32×10^3 L mo^{l-1} cm^{-1} at 470 nm What is the absorbance of a 6.24×10^{-5} M solution of the complex at 470 nm in a 1.00 cm cell? What is the percent transmittance of the solution described in (a)? What is the molar concentration of the complex in a solution that has the absorbance described in (a) when measured at 470 nm in a 5.00 cm cell?	05

<table>
<tr><td>3</td><td>294</td><td>The complex formed between Cu(I) and 1,10-phenanthroline has a molar absorptivity of 7000 $Lcm^{-1}mol^{-1}$ at 435 nm, the wavelength of maximum absorption. Calculate
a) The absorbance of a 6.77×10^{-5} M solution of a complex when measured in a 1.00 cm cell at 435 nm
b) The percent transmittance of a solution in (a)
c) The concentration of a solution in a 5.00 cm cell has the same absorbance as the solution in (a)
d) The path length through a 3.40×10^{-5} M solution of the complex that is needed for an absorbance that is the same as the solution in (a)</td><td>05</td></tr>
<tr><td>3</td><td>295</td><td>A compound X is to be determined by UV/visible spectrophotometry. A calibration curve is constructed from the solution of X with the following results: 0.50 ppm, A=0.24; 1.5 ppm, A=0.36; 2.5 ppm, A=0.44;3.5 ppm, A=0.59; 4.5 ppm, A=0.70. Find the slope and intercept of the calibration curve. Construct a plot of the calibration curve and determine the unknown concentration by hand from the plot.</td><td>05</td></tr>
<tr><td>3</td><td>296</td><td>One common way to determine phosphorus in urine is to treat the sample, after removing the protein, with molybdenum (VI) and then reduce the resulting 12-molybdophosphate complex with ascorbic acid to give an intense blue-colored species called molybdenum blue. The absorbance of the molybdenum can be measured at 650 nm. A patient produced 1122 ml of urine in 24 hrs. A 1.00 ml aliquot of the sample was treated with Mo(IV) and ascorbic acid and was diluted to a volume of 50.00 ml. The calibration curve was prepared by treating aliquots of phosphate standard solutions in the same manner as the urine sample. The absorbances of the standards and the urine sample were obtained at 650 nm, and the following results were obtained.
<table>
<tr><th>Solution</th><th></th></tr>
<tr><td>1.00 ppm P</td><td></td></tr>
<tr><td>2.00 ppm P</td><td></td></tr>
<tr><td>3.00 ppm P</td><td></td></tr>
<tr><td>4.00 ppm P</td><td></td></tr>
<tr><td>Urine sample</td><td></td></tr>
</table>
Find the slope, intercept. Construct a plot of a calibration curve. Determine the number of ppm of P in the urine sample. Compare the unknown concentration with that obtained manually from the calibration curve
What mass in grams of phosphorus was eliminated per day by the patient?
What is the phosphate concentration in urine in mM?</td><td>05</td></tr>
<tr><td>3</td><td>297</td><td>Nitrite is commonly determined by a colorimetric procedure using a reaction called the Griess reaction. In this reaction, the sample containing nitrite is reacted with sulphanilide and N-(1-naphthyl)ethylenediamine to form a colored species that absorbs at 550 nm. By using an automated flow analysis</td><td>05</td></tr>
</table>

<table>
<tr><td></td><td></td><td>instrument, the following results were obtained for standard solutions of nitrite and a sample containing an unknown amount. Construct a plot of the calibration curve
<table>
<tr><td>Solution (μM)</td><td></td></tr>
<tr><td>2.0</td><td></td></tr>
<tr><td>6.0</td><td></td></tr>
<tr><td>10.0</td><td></td></tr>
<tr><td>14.0</td><td></td></tr>
<tr><td>18.0</td><td></td></tr>
<tr><td>unknown</td><td></td></tr>
</table>
a) Find the slope, intercept of the calibration curve
b) Determine the concentration of nitrite in the sample</td><td></td></tr>
<tr><td>3</td><td>298</td><td>In the determination of acetone in biological fluids, the following calibration curve between absorbance and concentration was obtained.
<table>
<tr><td>ne standards (mg/100 ml)</td><td>bance</td></tr>
<tr><td>Reagent blank</td><td></td></tr>
<tr><td>0.5</td><td></td></tr>
<tr><td>1.0</td><td></td></tr>
<tr><td>2.0</td><td></td></tr>
<tr><td>4.0</td><td></td></tr>
<tr><td>6.0</td><td></td></tr>
<tr><td>8.0</td><td></td></tr>
</table>
Calculate the molar absorptivities and specific absorptivities</td><td>05</td></tr>
<tr><td>2</td><td>299</td><td>Deduce an expression for moment of inertia for linear triatomic molecules.</td><td>05</td></tr>
<tr><td>2</td><td>300</td><td>Explain the ‘electronic spectra of diatomic molecules.</td><td>05</td></tr>
<tr><td>2</td><td>301</td><td>Explain the rotation spectra of a rigid rotator.</td><td>05</td></tr>
<tr><td>2</td><td>302</td><td>Explain the rotation spectra of diatomic molecules.</td><td>05</td></tr>
<tr><td>2</td><td>303</td><td>Write a note on transitions observed in the rotational spectrum</td><td>05</td></tr>
<tr><td>2</td><td>304</td><td>Write a note on rotational energy levels and allowed spectral lines for rigid and non-rigid rotors.</td><td>05</td></tr>
<tr><td>2</td><td>305</td><td>Explain factors determining the intensity of spectral lines.</td><td>05</td></tr>
<tr><td>2</td><td>306</td><td>Explain the effect of isotopic substitution on rotation spectra.</td><td>05</td></tr>
<tr><td>2</td><td>307</td><td>What is the effect of isotopic substitution on microwave spectra of a linear diatomic molecule?</td><td>05</td></tr>
<tr><td>2</td><td>308</td><td>Write a brief note on the relative intensities of spectral lines.</td><td>05</td></tr>
<tr><td>2</td><td>309</td><td>Write a short note on factors affecting the intensity of spectral lines.</td><td>05</td></tr>
<tr><td>2</td><td>310</td><td>Write down the effect of isotopic substitution on the spectrum of carbon monoxide.</td><td>05</td></tr>
<tr><td>2</td><td>311</td><td>Explain the instrumentation technique involved in microwave spectrometer.</td><td>05</td></tr>
<tr><td>2</td><td>312</td><td>Obtain the expression for moment of inertia for a rigid diatomic molecule.</td><td>05</td></tr>
<tr><td>2</td><td>313</td><td>Obtain the expression for moment of inertia for a rigid diatomic molecule.</td><td>05</td></tr>
</table>

2	314	Distinguish between harmonic and anharmonic oscillator with respect to energy, selection rule, and zero-point energy	05
2	315	Explain simple harmonic oscillator and anharmonic oscillator and give selection rules.	05
2	316	Explain simple harmonic oscillator and anharmonic oscillator and give selection rules.	05
2	317	What is the significance of zero-point energy? Obtain an expression for zero-point energy of an anharmonic oscillator.	05
2	318	Explain the vibrational-rotational spectra of carbon monoxide	05
2	319	Obtain an expression for fundamental frequency and overtones for a simple harmonic oscillator.	05
2	320	What are parallel and perpendicular vibrations? Explain with an example.	05
2	321	What are symmetric and asymmetric vibrations? Explain with the example of H_2O molecule.	05
2	322	Distinguish between the symmetric top (prolate and oblate), spherical top, and asymmetric top molecules.	05
2	323	Explain different modes of vibrations for a linear triatomic molecule.	05
2	324	Calculate the allowed energy levels for rigid diatomic molecules and represent the resulting absorption spectrum.	05
2	325	The vibrational energy level $v=0$, $v=1$ of a diatomic molecule is separated by 2143 cm^{-1}. Its anharmonicity ($\omega_e x_e$) is 14 cm^{-1}. Calculate the ω_e and first overtone of this molecule.	05
2	326	Write a short note on P, Q, and R branches observed in the IR spectrum of a diatomic molecule.	05

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Question Paper Pattern- Model Question Paper

Q.P Code: 85322

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE			
III SEMESTER –COURSE M.SC FINAL EXAMINATION			
SUBJECT : CHEMISTRY			
TITLE: MOLECULAR SYMMETRY AND SPECTROSCOPY (HC)			
TIME: 3 HRS			**MAX MARKS: 70**
PART-A			
1		**Answer any ten of the following:**	**10x2=20**
	a	Identify the symmetry elements present in N_2H_2 (trans structure)	2
	b	Why does CO_2 possess an inversion centre, but NO_2 does not?	2
	c	List the groups which are the subgroups of D_{6h} and C_{3v}	2
	d	Identify the point group to which the following molecules belong Cis-butadiene, transbutadiene,	2

	e	For infrared radiation of 5.00 μm, what is the wave number in cm^{-1}	2
	f	Why is ethanol is a good solvent in ultraviolet	2
	g	Explain why it is necessary to use quartz cuvettes while recording UV spectra.	2
	h	Which of the following would show ESR spectrum? i) Methyl radical ii) Nitrogen molecule iii) Carbon di oxide iv) Benzene radical	2
	i	Give the classification of rigid rotors of polyatomic molecules.	2
	j	Write down the rule of mutual exclusion in Raman spectroscopy..	2
	k	Write the selection rule for the raman spectroscopy and rotational spectroscopy	2
	l	Write down selection rule for pure rotational Raman activity in linear molecules	2
PART-B			
		Answer any Five of the following:	5 x 10= 50
2	a	Prove that The readings 1,2,3,…..12 on the dial of a clock form a group under multiplication	5
	b	Determine the point group of PF_5 using the flowchart.	5
3	a	Construct the matrices for symmetry operations in the C_{2v} point group using H-1s orbitals and O-2pz orbital of water molecule (assume the z-axis to be C_2 axis). Write down the characters for this reducible representation and decompose in to its irreducible representation	5
	b	Find out the symmetries of the fundamentals of sulphate and carbonate. Comment on their infrared and Raman activities	5
4	a	The molar absorptivity of a particular solute is 2.1×10^4. Calculate the transmittance through a cuvette with 5.0 cm light path for a 2.0 μM solution.	5
	b	Biphenyl shows the following ultraviolet absorption data. In its 2,2'-dimethyl derivative the absorption pattern becomes almost similar to *O*-xylene. Why?	5
5	a	The UV Spectrum of acetone shows absorption bands at 166, 189 and 279 nm. What type of transition is responsible for each of these bands?	5
	b	Discuss the bathochromic shift of conjugated dienes based on HOMO-LUMO theory	5
6	a	Explain the rotation spectra of a rigid rotator.	5
	b	Explain factors determining the intensity of spectral lines.	5
7	a	Explain the effect of isotopic substitution on rotation spectra.	5
	b	Explain the rotation spectra of diatomic molecules.	5
8	a	Explain the vibrational-rotational spectra of carbon monoxide	5
	b	Discuss in detail the ESR spectrum of 1,4-benzoquinone radical.	5